

Fume and Gas Safety



SAFETY

CONTROLS

WIRE

TIG / STICK

WELDING TIPS

MAINTENANCE

INHALATION HAZARD:
Welding and Plasma Cutting produce toxic fumes.

- Exposure to welding or cutting exhaust fumes can increase the risk of developing certain cancers, such as cancer of the larynx and lung cancer.** Also, some diseases that may be linked to exposure to welding or plasma cutting exhaust fumes are:
 - Early onset of Parkinson's Disease
 - Heart disease
 - Ulcers
 - Damage to the reproductive organs
 - Inflammation of the small intestine or stomach
 - Kidney damage
 - Respiratory diseases such as emphysema, bronchitis, or pneumoniaUse natural or forced air ventilation and wear a respirator approved by NIOSH to protect against the fumes produced to reduce the risk of developing the above illnesses.
- Do not use near degreasing or painting operations.**
- Keep head out of fumes.**
Do not breathe exhaust fumes.
- Use enough ventilation, exhaust at arc, or both, to keep fumes and gases from breathing zone and general area.** If engineering controls are not feasible, use an approved respirator.
- Work in a confined area only if it is well-ventilated, or while wearing an air-supplied respirator.**
- Have a recognized specialist in Industrial Hygiene or Environmental Services check the operation and air quality and make recommendations for the specific welding situation.**
Follow OSHA guidelines for Permissible Exposure Limits (PEL's) and the American Conference of Governmental Industrial Hygienists recommendations for Threshold Limit Values (TLV's) for fumes and gases.

Arc Ray Safety



ARC RAYS can injure eyes and burn skin.

- Wear ANSI-approved welding eye protection featuring at least a number 10 shade lens rating.**
- Wear leather leggings, fire resistant shoes or boots during use.** Do not wear pants with cuffs, shirts with open pockets, or any clothing that can catch and hold molten metal or sparks.
- Keep clothing free of grease, oil, solvents, or any flammable substances.**
Wear dry, insulating gloves and protective clothing.
- Wear an approved head covering to protect the head and neck.** Use aprons, cape, sleeves, shoulder covers, and bibs designed and approved for welding and cutting procedures.
- Wear an approved welding jacket or long sleeves to protect forearms from radiation burns.**
- When welding/cutting overhead or in confined spaces, wear flame resistant ear plugs or ear muffs to keep sparks out of ears.**